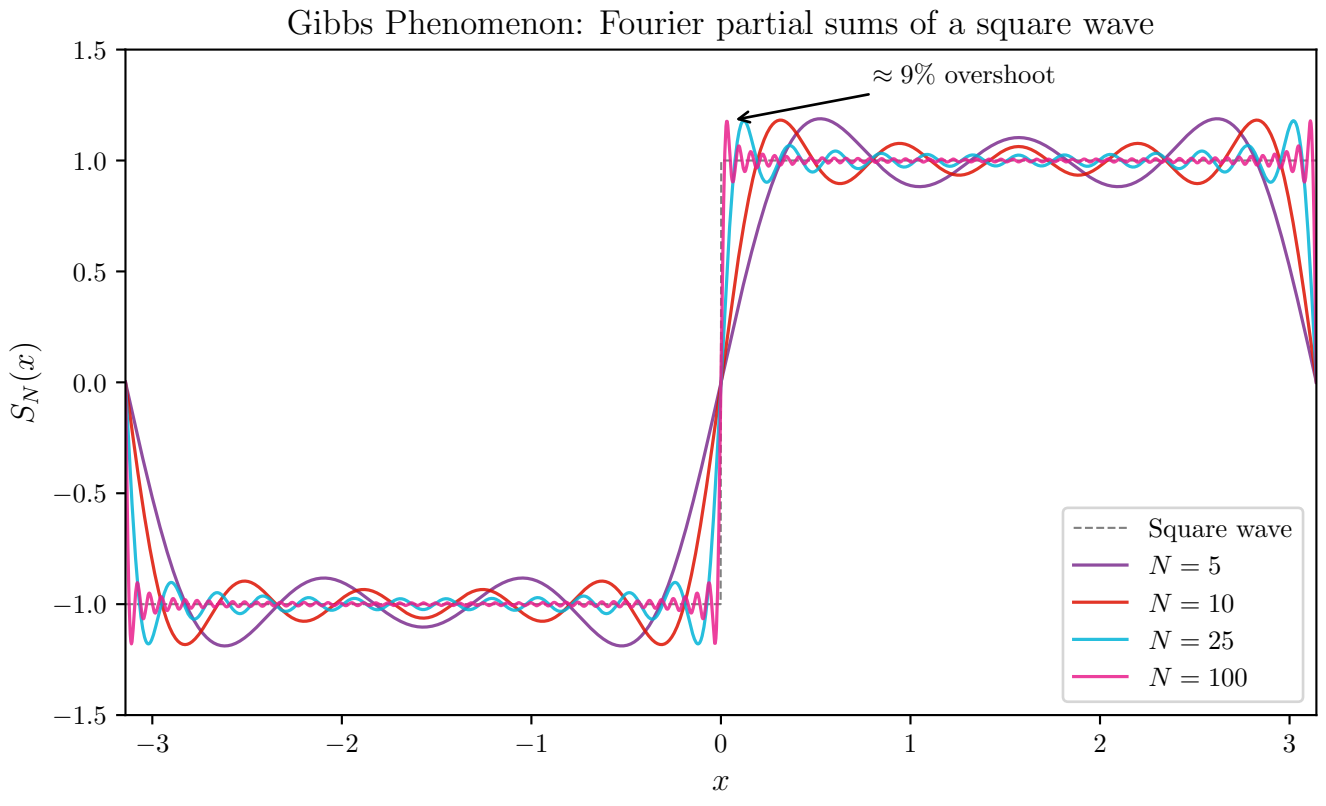
Gibbs Phenomenon: Fourier partial sums of a square wave
≈ 9% overshoot
Square wave
$N = 5$
$N = 10$
$N = 25$
$N = 100$
$S_N(x)$
x
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−3
−2
−1
0
1
2
3